

One-Way-ANOVA

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One Way ANOVA

Load data

```
data <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/One-Way-ANOVA/1")
data$Fertilizer <- as.factor(data$Fertilizer) # convert to factor so ANOVA treats it as categorical
```

Create ANOVA Model

To test whether mean yield differs among fertilizer groups

```
model <- aov(Yield ~ Fertilizer, data = data)
```

Assumption Checking

Shows one way anova is reliable or not.

1. Normality of Residuals

```
shapiro.test(residuals(model))
```

Shapiro-Wilk normality test

```
data: residuals(model)
W = 0.82816, p-value = 0.01994
```

Interpretation:

- $p > 0.05 \rightarrow$ residuals are approximately normal $\rightarrow$ acceptable
- $p < 0.05 \rightarrow$ deviation from normality $\rightarrow$ check plots before deciding

Understanding Normality

- Normal data $\rightarrow$ values are evenly distributed around mean. Example: (15, 13, 14, 16)
- Non-normal data $\rightarrow$ extreme variation/outliers. Example: (2, 5, 20, 50)

2. Homogeneity of Variance

```
library(car)
```

Warning: package 'car' was built under R version 4.4.3

Loading required package: carData

```
leveneTest(Yield ~ Fertilizer, data = data)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2      0      1
      9
```

Interpretation

1. P-Value is more than 0.05, the variances across groups are more or less equal (homogeneous), perform the one-way ANOVA reliably.
2. P-Value is less than 0.05 the variances across groups are not more or less equal (not homogeneous), use caution. see the plot

Homogeneity

For understanding homogeneity need to understand variance. Variance means the spread of data in each group.

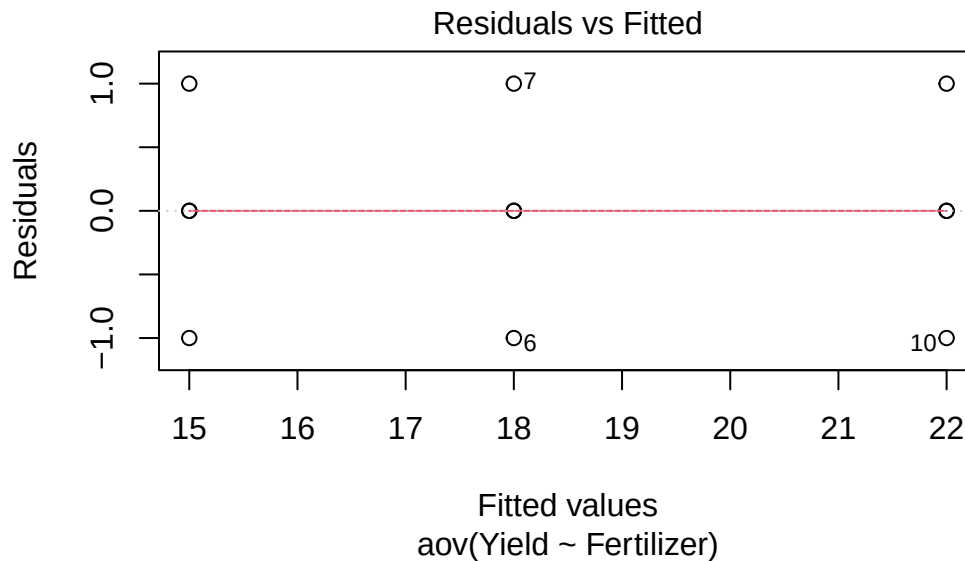
Example: Group A: (13, 14, 15, 16) → small spread → low variance

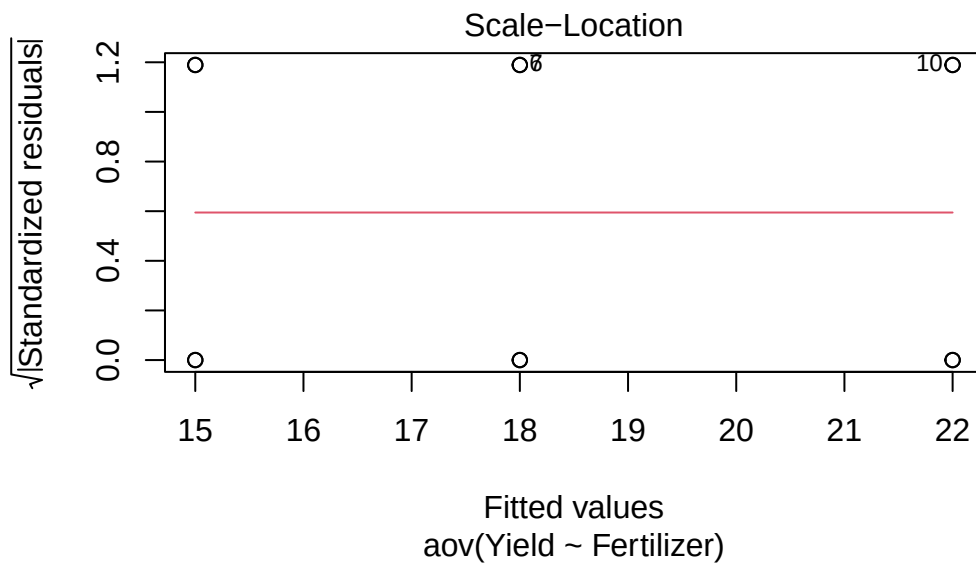
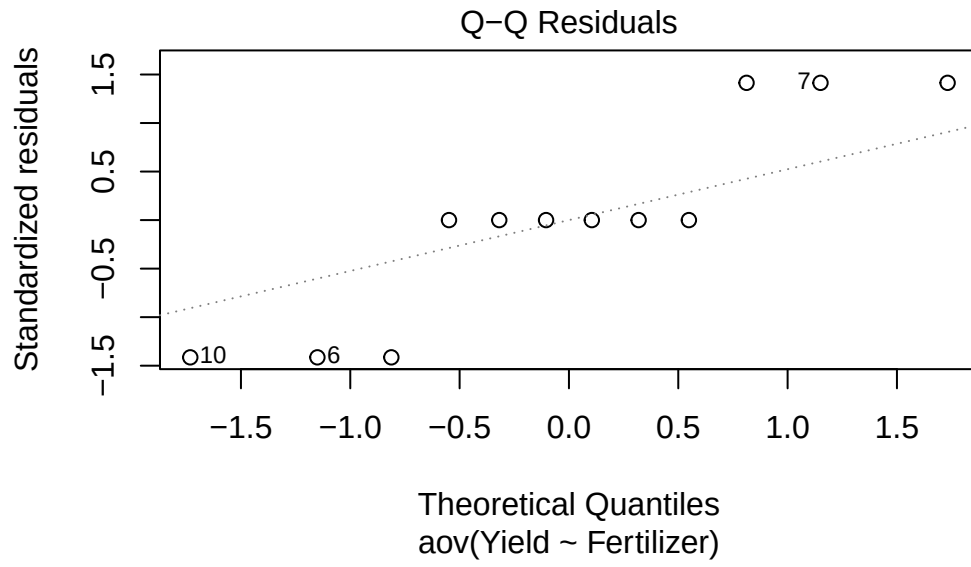
Group B: (10, 20, 30, 40) → large spread → high variance

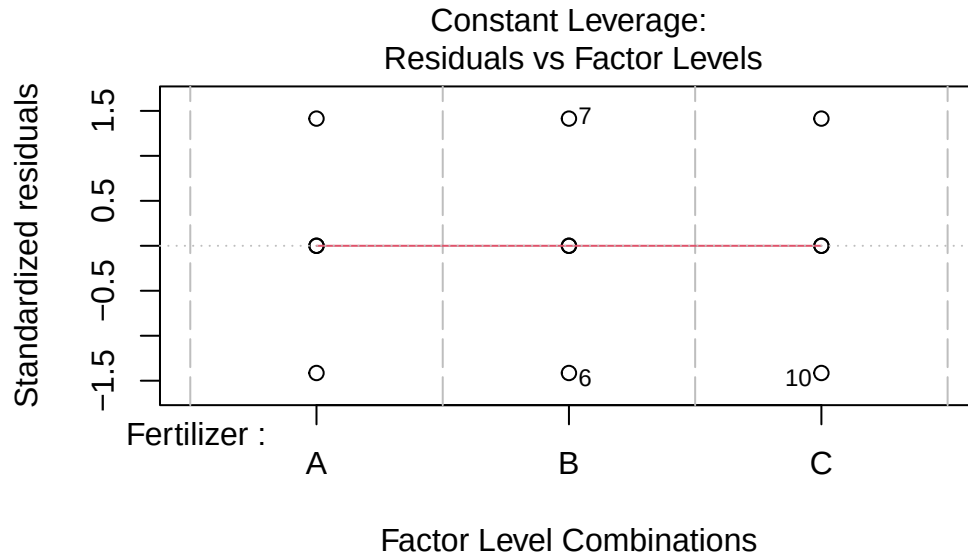
All groups have similar spread, the data is homogeneous. Ad one group is very spread out and others are tight, it's not homogeneous.

3. Diagnostic Plots

```
plot(model)
```







What to check:

Residuals vs Fitted → should show random pattern (no structure)

Normal Q-Q → points should lie near the straight line

Scale-Location → spread should be consistent

Residuals vs Leverage → detect outliers/influential points

These plots are more reliable than tests. Study more here!!!!

Perform anova

```
summary(model)
```

```

              Df Sum Sq Mean Sq F value    Pr(>F)
Fertilizer    2  98.67   49.33      74 2.59e-06 ***
Residuals     9    6.00    0.67
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Interpretation:

$p < 0.05 \rightarrow$ significant difference exists among groups

$p > 0.05 \rightarrow$ no strong evidence of difference

ANOVA tells “difference exists”. It does NOT tell which groups differ.

Alternative (Kruskal-Wallis Test)]**Use when**

- strong non-normality
- outliers present
- variance highly unequal

```
krus <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/One-Way-ANOVA/1  
kruskal.test(Yield ~ Fertilizer, data = krus)
```

Kruskal-Wallis rank sum test

data: Yield by Fertilizer

Kruskal-Wallis chi-squared = 4.3077, df = 2, p-value = 0.116

Interpretation:

$p < 0.05 \rightarrow$ at least one group differs

$p > 0.05 \rightarrow$ no evidence of difference